

MathGov Core v12.6 / SGP v8.5

Critical Disposition of the Claude Adversarial Audit

Current correction build: MathGov_Core_2026_09_v12.6_SGP_v8.5+2026.08.11.2

Audit source reviewed: MathGov_Core_v12_6_Adversarial_Audit.md

Disposition date: 11 August 2026

Semantic versions preserved: RippleLogic Canon v12.6; SGP v8.5; ripple.md v5.5; Agent System v12.4; CSV Gate Standard v2.4; Cascade Standard v2.6; WDBIP v1.6; Aligners Sheet v5.6; all other component semantic versions unchanged.

Executive verdict

The Claude audit contained a substantial amount of accurate specification analysis, particularly on the separation of qualification from ranking, the TRC formulation, Base/Welfare stream separation, the gate-critical confidence guard, non-dilution, non-maskability, SGP type separation, and scientific claim boundaries. Those positive findings reinforce the decision to preserve the five-stage architecture without redesign.

The audit also contained a decisive scope error. It explicitly examined only seventeen selected files and stated that no renderable DOCX, PDF, schema, validator, fixture, manifest, or release-verification surface had been supplied. That description did not match the complete publication-polished v12.6 package used as the source for this correction build. The latest package contains genuine OOXML DOCX files, publication PDFs, Markdown sources, schemas, validators, fixtures, manifests, hashes, QA records, and a complete master verifier. Therefore the audit's S0 format-integrity claim, stale-README claim, missing-machine-surface claim, and most release-engineering conclusions are not findings against the actual latest package.

After independent rechecking, the audit produced several genuine, bounded improvements. They have been implemented without changing the canonical cascade, equations, numerical defaults, rights-coverage sets, catastrophe-cell sets, SGP floors, or component semantic versions.

The final disposition is:

Accepted and implemented: sensitivity-induced non-decisiveness, Appendix B standalone completeness, explicit empty-set dispositions, non-default catastrophe-weight governance, post-saturation threshold-scale semantics, Emergency Mode clamp-tie ordering, Capability Claim Integrity discoverability, the Aligners Sheet CSV version correction, and exact TOC repagination.

Partially accepted and corrected more conservatively: D6 Meaning and local D7 Environment rights relevance. The misleading interpretive labels were corrected and protected effects must now be routed into the applicable canonical rights-covered dimensions or a governed extension. The audit's proposed wholesale expansion of DIGN and ECOL was not adopted because it would constitutionalize entire heterogeneous welfare dimensions without calibration.

Rejected: the false DOCX/PDF absence finding, stale README finding, missing-schema/validator finding, undeclared-MHIOS finding, fixed Delta_min = 0.02, full-calculation-on-load metadata as a substitute for live recalculation, and the claim that a new cascade graphic is essential.

Architecture preserved exactly

The governing cascade remains:

RG -> RF/NCRC -> TRC -> CSV -> RLS

Its functional structure remains:

Reality Grounding is the claim-authority precondition.

RF/NCRC, TRC, and CSV are the three hard admissibility/selectability gates.

RLS ranks only the selectable set.

UCI and HOI remain CSV-internal diagnostics and residual tie-break or monitoring surfaces where material.

No later function may compensate for or reinterpret failure at an earlier function.

No sixth gate, eighth welfare dimension, new right, new catastrophe category, new score, new authority layer, or external ontology was added.

Finding-by-finding disposition

Release-engineering findings

Audit finding

Disposition

Reason

All nominal DOCX files are plain text and no PDFs exist

Rejected as inapplicable to the latest package

The complete v12.6 build contains genuine OOXML DOCX files and publication PDFs. The audit's own scope declaration states that it reviewed a limited selected-file set and could not perform rendering or parity review.

README is entirely stale

Rejected for the latest package

The latest source README already identified v12.6 / SGP v8.5 and the current Core 15. This build advances only the immutable build identity and current audit link.

Schemas, validators, fixtures, manifests and CI are absent

Rejected for the latest package

These surfaces exist under schemas/, tests/, release/, and .github/workflows/. The master verifier exercises them.

MHIOS has no declared status

Rejected as a Core defect; clarified for discoverability

MHIOS is separately distributed as a pre-1.0 candidate human-interface and orchestration companion, not a Core 15 governing source. The README now states explicitly that the Canon controls in any conflict.

Canon TOC pagination could not be verified

Accepted as a final-QA task

The Canon's static linked TOC was remapped against the final 275-page PDF. All seventy entries now show the actual physical pages.

Rights-coverage findings

D6 Meaning

The audit correctly observed that Appendix AD used the phrase Rights Floor on selected D6 cells even though Appendix C.2 is the authoritative rights-coverage map and contains no direct D6 cell. That label could mislead a human reviewer into believing that D6 salience alone creates a non-compensatory floor.

The audit's proposed fix, extending DIGN to all D6 Meaning cells across U1-U6, was not adopted. D6 includes purpose, coherence, value congruence, identity, public meaning, and cultural continuity. Many of these can be important without being rights violations. Applying one threshold to the entire dimension would merge ordinary welfare, contested cultural evaluation, and legally or ethically protected interests into one broad constitutional category.

The implemented correction is narrower and more defensible:

Appendix AD now labels the six affected cells as Rights-relevance cue, not Rights Floor.

The Canon states that D6 does not acquire floor status merely because it is rights-relevant.

Where the underlying effect instantiates conscience, belief, coercion, discrimination, dignity, information access, due process, agency, or another protected interest, the run must construct and evidence the corresponding canonical rights-floor instance in the applicable covered dimension, or declare a governed rights-coverage extension.

WDBIP now states the same routing rule at the D6 boundary.

A material protected effect may not be left only as compensable D6 welfare.

This closes the interpretive contradiction without pretending that every meaning loss is categorically equivalent.

Local and household D7 Environment

The audit also correctly observed that selected local D7 cells were labelled Rights Floor although canonical ECOL coverage is not a blanket floor over every environmental cell. Again, the proposed expansion of ECOL to all U1-U7 D7 cells was not adopted. D7 includes air and water quality, local amenity, built-environment condition, biodiversity, resource cycles, environmental enabling systems, and many other states. A uniform direct rights threshold across all of them would create substantial uncalibrated overreach.

The implemented correction is:

The affected D7 cells are now Rights-relevance cue entries.

Environmental condition remains recorded in D7.

Health, habitability, basic-needs, discriminatory-burden, information, agency, due-process, dignity, and protected life-support consequences must be represented separately in the applicable canonical rights-covered dimensions or a governed extension.

A material protected effect may not be hidden as compensable environmental welfare.

WDBIP now makes this distinction explicit.

The canonical rights-coverage sets remain unchanged. This is deliberate: the correction improves routing and prevents underprotection without creating an empirically uncalibrated blanket right over every D6 or D7 impact.

Decisiveness and measurement resolution

This was the strongest substantive finding in the audit.

The worked-run workbook previously reported:

point-score Gap above the default delta = 2;

UNCERTAINTY_CALIBRATION_STATUS = UNVALIDATED;

sigma-times-two Gap below the decisiveness threshold;

UNCERTAINTY_DECISIVENESS_SENSITIVE;

yet RLS_DECISIVE = YES, ALLOW_FRAMEWORK_SELECTION, and no escalation.

That combination was internally overconfident. A computation that has already demonstrated that its decisive/non-decisive classification flips under a governed sensitivity may not ignore that result merely because the tier did not require the sensitivity calculation.

The following corrections were implemented:

Cross-tier computed-sensitivity rule. Whenever any tier actually computes a governed uncertainty, dependence, baseline, weighting, saturation, or related sensitivity and the decisive/non-decisive classification changes, the run must set the applicable sensitivity flag and be treated as non-decisive until tie-break or governance review is complete.

Measurement-resolution rule. Cell uncertainty must include material elicitation, measurement, coding, model, and anchor-resolution uncertainty. Zero is permitted only for exact deterministic input and transformation. Unsupported precision may not manufacture decisiveness through an arbitrarily small denominator.

Method C restriction. A bare unvalidated constant may support demonstrations, not an operational claim of unique framework selection.

Non-decisive enforcement. A sensitivity flip, unsupported uncertainty basis, or ordinary Gap failure blocks deterministic framework selection. Authority may still choose among selectable options, but the record must identify that as authority selection.

Validator hardening. A run record claiming decisive = true while declaring hypothesis-sensitive or parameter-sensitive ranking is now rejected.

New adversarial fixture. fail_sensitive_ranking_claims_decisive.json proves the rejection path.

Worked-run correction. The Aligners Sheet now reports REFUSE_DETERMINISTIC_SELECTION, identifies A only as the point-score leader, records ESCALATION_REQUIRED = YES, and separately records the accountable authority's selection of A.

Why a fixed Delta_min was not added

The audit proposed Delta_min = 0.02. The underlying concern is valid, but the proposed value is not yet empirically calibrated. A universal absolute floor can create its own false precision, ignore domain-specific elicitation resolution, and double-count uncertainty already represented through sigma. The implemented measurement-resolution rule addresses the real defect while leaving future empirical work to calibrate whether a portable absolute floor is justified.

Other accepted specification corrections

Appendix B standalone completeness

Appendix B describes itself as the Tier 1-3 executable equation pack. It now reproduces the already-governing rules needed for standalone implementation:

empty protected-subgroup-set disposition;

$Q = 0 \rightarrow \text{RLS_NO_ACTIVE_MASS};$

dependence-cluster stress and measurement-resolution requirements;

sensitivity-induced non-decisiveness;

empty containment-map escalation;

all-NA UCI $\rightarrow$ UNAVAILABLE.

No new formula was introduced. Existing controlling rules were restored to the surface that claims standalone executability.

Emergency Mode clamp ties

The conservative subgroup bound remains clamped to $[-1, +1]$ for ordinary admissibility. Where multiple Emergency Mode options tie because the clamp maps all of them to -1, the unclamped conservative value is now used only as a secondary ordering key. This preserves the canonical floor while restoring least-bad discrimination under emergency comparison. It never converts a failing option into an ordinary NCRC pass.

Non-default catastrophe weights

Any non-default catastrophe-weight vector must now be:

declared before outcome inspection;

normalized within the active profile;

justified by a named source;

applied symmetrically;

consistent with non-dilution;

accompanied, for Tier 3 and decision-material or high-stakes Tier 2 use, by a uniform-weight counterfactual and sensitivity disclosure.

This closes a genuine governance gap without changing the default uniform weights or the CVaR equation.

Post-saturation threshold scale

The Canon now states expressly that rights thresholds live on the post-saturation I_rights scale. It also states that $\tanh(\text{beta} * x)$ with the canonical $\text{beta} = 2$ expands small magnitudes near zero and compresses near the bounds; it is a bounded shaping transform, not a small-signal attenuator. Threshold calibration and sensitivity review must identify the scale used.

Empty-set dispositions

Three potentially ambiguous edge cases now fail closed:

an empty protected-subgroup set cannot create a vacuous rights pass;

an empty `Contain(u,G_c)` set for a triggered scope produces `CONTAINMENT_MAP_INCOMPLETE` and escalation or unresolved CSV status;

an empty applicable UCI component set produces `UCI_UNAVAILABLE`, with no imputation.

Capability Claim Integrity discoverability

The portable `ripple.md` surface and SGP now use the canonical phrase Capability Claim Integrity and cross-reference Canon Section 2.1C. The rule itself was already correct; the change improves audit and implementation discoverability.

Gate numbering

The Canon now states explicitly that Formal gate 1-3 counts only RF/NCRC, TRC, and CSV. RG is the claim-authority precondition, and RLS is the ranking stage. This removes a possible fast-reader ambiguity without changing the public five-level method.

Workbook corrections

The Aligners Sheet remains v5.6. Its formulas, sheet inventory, and semantic role are unchanged.

Implemented corrections include:

`Core_Component_Map` now pins CSV Gate Standard v2.4, not v2.3.

`RLS_DECISIVE` = NO.

`FRAMEWORK_VERDICT` = `REFUSE_DETERMINISTIC_SELECTION`.

A remains the point-score leader, not a unique framework selection.

A separate `AuthoritySelectionRecord` identifies the accountable authority's selection of A.

`ESCALATION_REQUIRED` = YES.

`UNCERTAINTY_DECISIVENESS_SENSITIVE` = YES remains visible rather than being neutralized.

Dashboard, Display Clarity, PCC, Audit Flags, Verdict Hardening, Sanity Checklist, PLSS, Qualification Continuity, and release-note surfaces were synchronized.

The workbook and Core 15 mirror remain byte-identical.

The workbook still contains 87 worksheets and 1,643 formulas. Independent isolated LibreOffice recalculation returns zero formula errors and preserves the qualification-continuity record.

Rejected or deferred recommendations

FullCalcOnLoad metadata

Not implemented. The current workbook is independently recalculated in an isolated office profile during release verification. That is stronger evidence than merely instructing a future reader's application to recalculate on opening. The live-recalculation verifier passes with 1,643 formulas and zero errors.

New essential cascade graphic

Not implemented. The package already includes a dedicated Cascade Standard, clear layer tables, a qualification-lifecycle asset, the 3R/1-2 teaching structure, and linked navigation. The accepted corrections do not create a new figure-dependent concept. Adding another figure would duplicate existing explanatory surfaces and create synchronization and accessibility obligations without a demonstrated comprehension gap.

Broad semantic-version change

Not implemented. The five-stage architecture, formulas, numerical defaults, canonical rights sets, catastrophe sets, SGP floors, and component interfaces remain intact. The changes correct overclaim, routing, standalone completeness, edge-case behavior, and release consistency. They are issued as immutable build 2026.08.11.2 under the same semantic versions requested by the framework owner.

MHIOS content revision

Not implemented because no defect in MHIOS v0.7 was established by this audit. The complete MHIOS package was not the object actually analysed in the selected-file audit. Its separate candidate status and Canon-subordination boundary are now stated more visibly in the Core README.

Exact changed files

Governing and companion sources

docs/canon/RippleLogic_v12.6_Canon.md/.docx/.pdf

docs/standards/wdbip/Welfare_Dimension_Boundary_and_Interaction_Protocol_v1.6.md/.docx/.pdf

docs/standards/ripple_md_Standard_v5.5.md/.docx/.pdf

docs/sgp/SGP_v8.5.md/.docx/.pdf

docs/canon/AD_49_Cell_Welfare_Dictionary.csv

Machine and workbook surfaces

release/VALIDATE_MATHGOV_RUN.py

tests/run_records/fail_sensitive_ranking_claims_decisive.json

schemas/mathgov_run_record_v4.schema.json for exact build binding

all active run records for exact build binding

docs/aligners/RippleLogic_Aligners_Sheet_v5.6.xlsx

core_15/RippleLogic_Aligners_Sheet_v5.6.xlsx

Release and assurance surfaces

VERSION_MANIFEST.yaml

README.md, START_HERE.md, START_HERE_RELEASE_INDEX_v12.6.md

VALIDATION_STATUS.md, VERIFICATION_LOG.md, RELEASE_CLAIMS_AND_NON_CLAIMS.md

release-reality registry, manifests, hashes, QA reports, verification report, release notes, current pin verifier, reality-coherence verifier, and master verifier.

Publication, table, and rendering audit

The final current Core publication set contains:

Metric

Result

Primary Markdown/DOCX/PDF triples

15

PDF pages

598

Tables

287

Table rows

2,439

Headings

1,634

Static linked TOC entries

125

Comments

0

Tracked changes

0

Content controls

0

Confirmed page-boundary failures

0

Tiny-font spans under 6 pt

0

Accessibility findings

0 high / 0 medium / 76 low raw-reference URLs

Every changed publication surface was rendered after final edits:

Canon: 275 pages;

SGP: 55 pages;

WDBIP: 34 pages;

ripple.md Standard: 66 pages.

All 430 changed pages were inspected through full-page contact review, with changed and high-risk pages inspected at full size. The Canon TOC was then repaginated against the final 275-page PDF and re-rendered. No clipped text, overlap, unreadable table, broken symbol, malformed footer, unintended blank page, or confirmed boundary failure remains.

The table system remains unchanged because it was already professionally complete: explicit borders, restrained light-blue headers, adequate cell padding, wrapped text, and repeating header rows. No table required structural redesign. The six changed Appendix AD rows were reviewed at full size and remain clear and legible.

Verification summary

Verification surface

Final result

Current pins and exact build identity

PASS

Nine subordinate release-verification surfaces

PASS

Core run records

6 positive / 30 expected failures / 1 active example

SGP RMCP

1 positive / 1 expected failure

WDBIP

1 positive / 17 expected failures

Aligners Sheet

87 sheets / 1,643 formulas / 0 formula errors

Primary reading mirrors

15 triples / 598 pages

Core 15 OOXML integrity

PASS

Static linked TOCs

125 entries; Canon pages remapped to final rendering

Accessibility

PASS: 0 high / 0 medium

SHA-256 active inventory and Core 15 ledger

PASS

Clean ZIP extraction and complete verifier replay

PASS

Final scientific boundary

This build establishes internal specification consistency, artifact integrity, machine-interface conformance for the tested surfaces, workbook formula integrity, and publication rendering quality.

It does not establish:

construct validity;
inter-rater reliability;
empirical superiority;
legal authority;
physical-safety certification;
consciousness or sentience detection;
complete moral truth;
institutional legitimacy in a specific jurisdiction;
deployment readiness.

Final disposition

The Claude audit was useful but not uniformly accurate. Its strongest technical finding, decisiveness under sensitivity, has been corrected. Its Appendix B, edge-case, catastrophe-weight, threshold-scale, cross-reference, and workbook-pin findings were also valuable. Its release-engineering claims were based on an incomplete selected-file set rather than the latest complete package. Its proposed broad D6/D7 rights expansions and fixed absolute decisiveness floor exceeded the available calibration evidence and were not adopted.

MathGov Core v12.6 / SGP v8.5 build 2026.08.11.2 is complete and ready as an internally verified, publication-polished research specification. The architecture should now remain frozen unless empirical work, external review of the complete package, or a genuinely new requirement identifies a substantive defect.